

College of Business Administration

Factors affecting the intention to use artificial intelligence in mobile banking

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Background and problem statement

Technology plays a major role in our daily lives, and it is hard to imagine life without it. Among the most influential technologies are artificial intelligence (AI) and smart devices, which have transformed many services—including mobile banking (m-banking). M-banking allows users to perform financial transactions such as paying bills and transferring money anytime and anywhere. However, challenges like fear of fraud, lack of trust, and limited access to technology still hinder the adoption of AI in banking. This study aims to understand customer behavior, enhance user trust and satisfaction, and reduce concerns about AI in order to increase the adoption of mobile banking services.

AI adoption in mobile banking faces key challenges such as privacy concerns, low user trust, and system complexity. Customer hesitation also stems from lack of awareness and security fears. This study explores these barriers and suggests solutions for a safer, smoother user experience.

Research Questions and objectives

Based on the above discussion, this study sought to answer the following questions:
What are the factors that affect Intention to use artificial intelligence in mobile banking?
To what extent does each factor affect the adoption of artificial intelligence in banking applications from the users’ perspective?

This study aims to explore the factors influencing users’ intention to adopt AI in mobile banking by:

- Reviewing relevant literature to build a strong theoretical foundation
- Developing and testing a research model to identify key influencing factors.

Methods and Materials

This study uses a descriptive and quantitative research design to explore factors influencing the intention to use AI in mobile banking among students at Sattam bin Abdulaziz University. Data was collected using a questionnaire, developed based on recent literature. The quantitative approach allowed for statistical analysis of real data to support the research objectives.

Results

- Trust and security come first, followed by user experience, then perceived benefits.
- Younger, more educated users are more inclined to adopt AI services, and males tend to trust AI security more than females.
- The success of promoting AI banking services heavily depends on building trust and ensuring ease of use.
- Strengths and limitations of methodology

Conclusion

This study identified key factors influencing the adoption of AI in banking services, including trust, user satisfaction, perceived ease of use, and actual user behavior. Trust emerged as the most critical factor—without it, users are unlikely to engage with AI features. Satisfaction and ease of use were also essential; users prefer systems that are intuitive and meet their expectations. To encourage adoption, banks must offer secure, user-friendly platforms and bridge the gap between user intention and actual behavior. Addressing challenges like data privacy, user education, and system reliability is crucial. Recommendation: Building strong trust should be the top priority. Financial institutions must ensure transparency, robust security, and consistent service quality to gain user confidence, drive adoption, and stay competitive in the evolving AI landscape.

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